



BrightLearn Sales Data Engineering

END TO END ETL PIPELINE USING SQL SERVER,
SSIS AND A STAR SCHEMA DATA WAREHOUSE

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Overview

The objective of this project was to transform raw sales data into a structured data warehouse that supports business reporting and decision making.

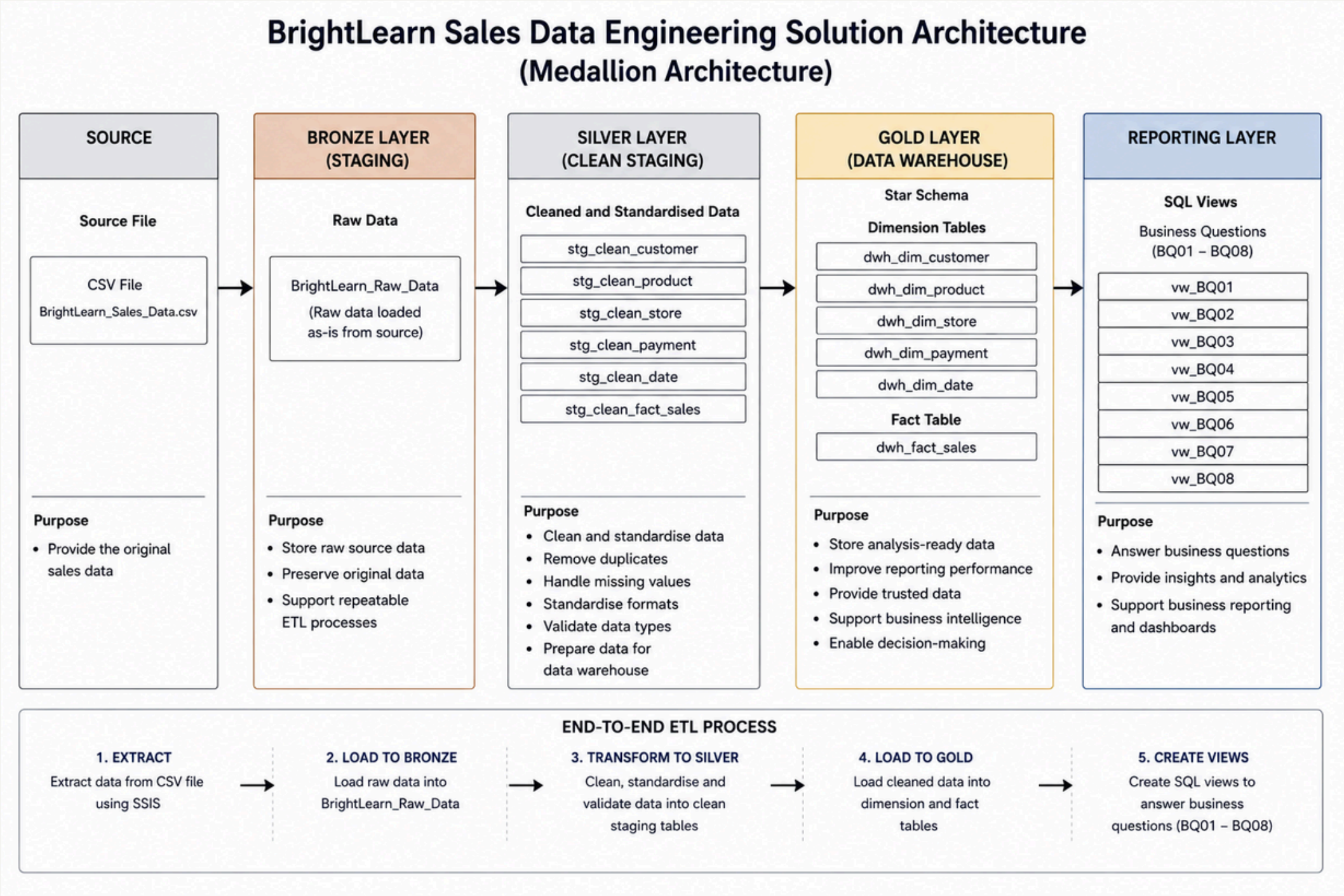
The solution includes:

- Importing data from a CSV file
- Building an automated ETL pipeline
- Cleaning and transforming data
- Designing a Star Schema data warehouse
- Creating SQL views to answer business questions

Business Problem

BrightLearn is a South African retail chain operating five stores across the country. The company stores all POS transaction data in a single raw CSV file containing customer, product, store and sales information in one denormalised table. The data also contains quality issues such as duplicate records, missing values, inconsistent formatting, multiple date formats and incorrect data types, making it difficult for the Business Intelligence team to produce reliable reports. As a result, management could not make informed decisions regarding inventory, customer retention and store performance. To address these challenges, the business required a data warehouse that cleans, organises and integrates the data into a structured and reliable source for reporting and analysis

Project Architecture



ETL Process

ETL Workflow

1. Extract

Imported the BrightLearn CSV file into SQL Server using SSIS.

2. Transform

Performed data cleaning including:

- Removing duplicates
- Trimming spaces
- Converting text to lowercase
- Standardising phone numbers
- Standardising date formats
- Handling missing values
- Converting numeric and monetary columns
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3. Load

Loaded clean data into the Star Schema data warehouse.

Data Warehouse Design

Dimension tables

- Customer
- Product
- Store
- Payment
- Date

Fact table

Sales Fact

Why a Star Schema was chosen:

- Simpler structure
- Faster queries
- Easy reporting
- Widely used in BI solutions

Business Questions

SQL Views (BQ01–BQ08) were created to answer the reporting requirements.

The questions included:

- Best selling products
- Revenue by store
- Month over month growth
- Top loyalty customers
- Win back campaign customers
- Average transaction value
- Product category performance
- Inventory below reorder level

Key Findings

- Kitchen Knife Set generated the highest revenue.
- BrightMart Sandton City achieved the highest sales.
- Revenue increased by 37.01% from May to June.
- Gold customers dominated the top spenders.
- Silver customers recorded the highest average transaction value.
- Groceries sold the highest quantity.
- No customers qualified for the Win-Back campaign.
- No products were below the reorder threshold in June.

Technologies used

- Microsoft SQL Server (SSMS)
- SQL Server Integration Services (SSIS)
- Transact-SQL (T-SQL)
- Visual Studio
- Git
- GitHub
- CSV Dataset

Lessons learned

- Importance of data quality
- Practical ETL development
- Data warehouse design
- SQL optimisation
- Automation using SSIS

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